# Workshop: Two-Way ANOVA

your name here

This week, we continue working with the Totally Serious Pop Culture Research Institute™ on research questions in Equestria.

Let's call in the packages and data we need,

```
library(tidyverse)
library(ssstats)
applebuck_recovery       <-       read_csv("https://raw.githubusercontent.com/
samanthaseals/STA4173Fa26/refs/heads/main/workshops/data/M2-2-applebuck_
recovery.csv")
crystal_focus <- read_csv("https://raw.githubusercontent.com/samanthaseals/STA
4173Fa26/refs/heads/main/workshops/data/M2-2-crystal_focus.csv")
friendship_learning      <-       read_csv("https://raw.githubusercontent.com/
samanthaseals/STA4173Fa26/refs/heads/main/workshops/data/M2-2-friendship_
learning.csv")
```

## Study 1: Crystal Focus

The Totally Serious Pop Culture Research Institute™ has opened an Equestria branch! Twilight Sparkle conducted an experiment to determine whether preparation method and pony type are associated with focus during a crystal-focus task. Researchers recruited 90 ponies: 30 Earth ponies, 30 pegasi, and 30 unicorns. Within each pony type, 15 ponies were randomly assigned to quiet practice and 15 to guided visualization.

After preparing, each pony completed the same task and received a focus score from 0 to 100, with higher scores indicating better focus. The researchers want to determine whether the effect of preparation method on average focus scores depends on pony type.

Dataset name: `crystal_focus`

Variables:

- `pony_id`: unique pony identifier.
- `pony_type`: Earth pony, Pegasus, or Unicorn.
- `preparation`: assigned condition.
- `focus_score`: response described above.

1a. Find the summary statistics to describe this data for each combination of the two explanatory variables.

```
crystal_focus |> posthoc(outcome = focus_score,
                         A = preparation,
                         B = pony_type)
```

```
Post-hoc analysis: Tukey's HSD

Following: two-way ANOVA (focus_score ~ preparation * pony_type)

Main effect: preparation
Comparison                                Difference  p-value
──────────────────────────────────────────────────────────────
Quiet practice vs Guided visualization      1.1444    0.282
──────────────────────────────────────────────────────────────


Main effect: pony_type
Comparison             Difference  p-value
──────────────────────────────────────────
Pegasus vs Earth pony     -1.2800    0.586
Unicorn vs Earth pony     -0.2400    0.981
Unicorn vs Pegasus         1.0400    0.702
──────────────────────────────────────────


Interaction: preparation × pony_type
Comparison                                                    Difference  p-
value
──────────────────────────────────────────────────────────────────────────────
Quiet practice:Earth pony vs Guided visualization:Earth pony       -21.1267  <
0.001  *
Guided visualization:Pegasus vs Guided visualization:Earth pony    -12.5667  <
0.001  *
Quiet practice:Pegasus vs Guided visualization:Earth pony          -11.1200  <
0.001  *
Guided visualization:Unicorn vs Guided visualization:Earth pony    -22.3600  <
0.001  *
Quiet practice:Unicorn vs Guided visualization:Earth pony             0.7533
0.998
Guided visualization:Pegasus vs Quiet practice:Earth pony             8.5600  <
0.001  *
Quiet practice:Pegasus vs Quiet practice:Earth pony                  10.0067  <
0.001  *
Guided visualization:Unicorn vs Quiet practice:Earth pony            -1.2333
0.984
Quiet practice:Unicorn vs Quiet practice:Earth pony                  21.8800  <
0.001  *
Quiet practice:Pegasus vs Guided visualization:Pegasus                1.4467
0.969
Guided visualization:Unicorn vs Guided visualization:Pegasus         -9.7933  <
```

```
0.001  *
Quiet practice:Unicorn vs Guided visualization:Pegasus              13.3200  <
0.001  *
Guided visualization:Unicorn vs Quiet practice:Pegasus             -11.2400  <
0.001  *
Quiet practice:Unicorn vs Quiet practice:Pegasus                    11.8733  <
0.001  *
Quiet practice:Unicorn vs Guided visualization:Unicorn              23.1133  <
0.001  *
-------------------------------------------------------------------------------


* p < α = 0.050
```

1b. Construct the appropriate two-way ANOVA table to determine if there is an interaction between preparation and pony type.

1c. Conduct the appropriate hypothesis test to show that there is an interaction effect.

**Hypotheses**:

- $H_0$ :
- $H_1$ :

**Test Statistic and $p$-value**

- $F =$
- Degrees of freedom:
- $p =$

**Rejection Region**

- Reject $H_0$ if ...

**Conclusion and Interpretation**

- Reject or fail to reject $H_0$.
- There is or is not sufficient evidence to conclude...

1d. Construct the appropriate posthoc test for a confirmatory study.

1e. Construct a profile plot for this data.

1f. Write a few sentences to describe what's going on with the data using the results of the ANOVA, the posthoc testing, and the profile plot above.

Your answer here.

# Study 2: School of Friendship

The Totally Serious Pop Culture Research Institute worked with Starlight Glimmer to compare lesson formats at the School of Friendship. Researchers recruited 90 ponies: 30 Earth ponies, 30

pegasi, and 30 unicorns. Within each pony type, 15 ponies were randomly assigned to an individual lesson and 15 to a cooperative lesson.

After the lesson, each pony completed the same assessment scored from 0 to 100, with higher scores indicating greater learning. The researchers want to determine whether the effect of lesson format on average learning scores depends on pony type.

Dataset name: `friendship_learning`

Variables:

- `pony_id`: unique pony identifier.
- `pony_type`: Earth pony, Pegasus, or Unicorn.
- `lesson_format`: assigned condition.
- `learning_score`: response described above.

2a. Find the summary statistics to describe this data for each combination of the two explanatory variables.

2b. Construct the appropriate two-way ANOVA table to determine if there is an interaction between preparation and pony type.

2c. Conduct the appropriate hypothesis test to show that there is NOT an interaction effect.

**Hypotheses**:

- $H_0$ :
- $H_1$ :

**Test Statistic and $p$-value**

- $F =$
- Degrees of freedom:
- $p =$

**Rejection Region**

- Reject $H_0$ if ...

**Conclusion and Interpretation**

- Reject or fail to reject $H_0$.
- There is or is not sufficient evidence to conclude...

2d. Reconstruct the two-way ANOVA table without the interaction term.

2e. Conduct the appropriate hypothesis test to determine if there is a relationship between learning score and pony type.

**Hypotheses**:

- $H_0$ :
- $H_1$ :

**Test Statistic and $p$-value**

- $F =$
- Degrees of freedom:
- $p =$

**Rejection Region**

- Reject $H_0$ if ...

**Conclusion and Interpretation**

- Reject or fail to reject $H_0$.
- There is or is not sufficient evidence to conclude...

2f. Conduct the appropriate hypothesis test to determine if there is a relationship between learning score and lesson format.

**Hypotheses**:

- $H_0$ :
- $H_1$ :

**Test Statistic and $p$-value**

- $F =$
- Degrees of freedom:
- $p =$

**Rejection Region**

- Reject $H_0$ if ...

**Conclusion and Interpretation**

- Reject or fail to reject $H_0$.
- There is or is not sufficient evidence to conclude...

2g. Construct the appropriate posthoc tests for an exploratory study.

2h. Construct a profile plot for this data.

2i. Write a few sentences to describe what's going on with the data using the results of the ANOVA, the posthoc testing, and the profile plot above.

Your answer here.

# Study 3: Sweet Apple Acres

The Totally Serious Pop Culture Research Institute partnered with Applejack to investigate recovery after applebucking. Researchers recruited 90 ponies: 30 Earth ponies, 30 pegasi, and 30 unicorns. After a standardized applebucking session, 15 ponies within each pony type were randomly assigned to rest and 15 to stretching.

Researchers recorded the number of minutes until each pony returned to its pre-session heart rate, with lower values indicating faster recovery. The researchers want to determine whether the effect of recovery routine on average recovery time depends on pony type.

Dataset name: `applebuck_recovery`

Variables:

- `pony_id`: unique pony identifier.
- `pony_type`: Earth pony, Pegasus, or Unicorn.
- `recovery_routine`: assigned condition.
- `recovery_minutes`: response described above.

3a. Find the summary statistics to describe this data for each combination of the two explanatory variables.

3b. Construct the appropriate two-way ANOVA table to determine if there is an interaction between recovery routine and pony type.

3c. Conduct the appropriate hypothesis test to show that there is an interaction effect.

**Hypotheses**:

- $H_0$ :
- $H_1$ :

**Test Statistic and $p$-value**

- $F =$
- Degrees of freedom:
- $p =$

**Rejection Region**

- Reject $H_0$ if ...

**Conclusion and Interpretation**

- Reject or fail to reject $H_0$.
- There is or is not sufficient evidence to conclude...

3d. Construct the appropriate posthoc test for a confirmatory study.

3e. Construct a profile plot for this data.

3f. Write a few sentences to describe what's going on with the data using the results of the ANOVA, the posthoc testing, and the profile plot above.

Your answer here.